Atul Kumar Tiwari

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Professional Summary

AI-enabled Full-Stack Developer with 4+ years of experience building scalable **Python-based web** and **GenAI applications** using **React.js, Next.js, FastAPI, and Flask**. Strong expertise in **LangChain-based RAG** systems, vector databases, and Azure AI services, with proven experience designing, deploying, and scaling enterprise-grade AI solutions on Azure App Services.

Technical Skills

- **Programming Languages:** Python, JavaScript, PHP, SQL, C++, HTML5, CSS3
- **Frontend:** React.js, Next.js, Bootstrap , Tailwind CSS, SASS/SCSS, jQuery
- **Backend & APIs:** FastAPI, Flask, Express.js, REST APIs
- **AI / GenAI & RAG**: LangChain, Azure OpenAI, Prompt Engineering, Embeddings, RAG Pipelines, Vector Search, Semantic Search, Document Chunking
- **RAG Patterns**: Naive RAG, Hybrid Search RAG, Conversational RAG
- **Vector Databases:** FAISS, ChromaDB
- **Databases:** MySQL, MongoDB, Azure Cosmos DB
- **Cloud & DevOps:** Azure App Services, Azure AI Search, Azure Document Intelligence, Azure Model Foundry, Git, GitHub, Jenkins, Postman, Docker (basic)

Work Experience

Software Developer – Noventiq India

February 2025 - Present

Achievements and Responsibilities:

- Designed and implemented **LangChain-based RAG pipelines** using **Azure OpenAI embeddings** with **FAISS and ChromaDB** for semantic document retrieval.

- Integrated **Azure Document Intelligence** to extract structured data from PDFs and scanned documents.
- Developed and deployed **Python-based FastAPI microservices** on **Azure App Services** for AI and document-processing workflows.
- Implemented **Azure AI Search with Vector Search** to build enterprise-grade intelligent search systems.
- Built scalable and responsive frontend modules using **React.js, Next.js, and Tailwind CSS**.
- Collaborated with cross-functional teams to deliver **end-to-end AI-powered web applications**.
- Optimized RAG pipelines by tuning chunk size, top-k retrieval, and similarity thresholds to improve answer relevance.

Web Developer - Crimson Interactive

January 2023 –February 2025

Achievements and Responsibilities:

- Developed 50+ responsive web pages using HTML, CSS, JavaScript,React.js, Next.js, and Flask, improving user engagement by 20%.
- Managed and optimized databases, resulting in a 15% reduction in data retrieval times.
- Implemented backend functionalities using Python (Flask), enhancing system performance by 25%.
- Optimized core web vitals, achieving a 30% increase in page load speed.
- Worked on **cloud-hosted applications deployed on Azure App Services**, supporting scalable and reliable web delivery.
- Integrated **Azure Blob Storage** for managing static assets, documents, and media files used across web applications.
- Assisted in implementing **Azure SQL / Azure Cosmos DB (basic usage)** for structured and semistructured data storage.
- Collaborated on **document upload and processing workflows**, laying the foundation for later adoption of **Azure Document Intelligence**.

Frontend Developer - Blue Meteor Pvt Ltd.

March 2022 – August 2022

Achievements and Responsibilities:

- Developed user interfaces with HTML5, CSS3, SCSS, JavaScript, and jQuery, leading to a 15% increase in user satisfaction.
- Implemented page optimization techniques, reducing load times by 20%.
- Ensured cross-browser compatibility, decreasing user-reported issues by 10%.
- Collaborated with a team of 5 to introduce new features, enhancing product functionality.

Web UI Developer - Realty Quarter Pvt Ltd.

June 2020 – March 2022

Achievements and Responsibilities:

- Implemented responsive layouts across various devices, boosting mobile user engagement by 30%.
- Utilized SASS and LESS preprocessors to streamline CSS development, reducing code redundancy by 20%.
- Tested and debugged websites, decreasing bug-related downtimes by 15%.
- Maintained front-end components, ensuring a 99% uptime for client websites.

AI / GenAI Projects

Intelligent Document Search System (RAG):

- Built a **LangChain-based RAG system** for contextual question answering over enterprise documents.
- Implemented **document ingestion, chunking, and vector indexing** using **Azure OpenAI embeddings** with **FAISS** and **ChromaDB**.
- Integrated **Azure Document Intelligence** to extract structured data from PDFs and scanned documents.
- Enabled **semantic and vector-based search** using **Azure AI Search (Vector Search)**.
- Developed backend APIs using **FastAPI** and frontend using **Next.js**.
- Deployed the end-to-end solution on **Azure App Services**.

Enterprise Knowledge Base Chatbot (Hybrid RAG):

- Designed a **multi-document RAG chatbot** for internal enterprise knowledge (HR policies, SOPs, technical docs).
- Implemented **hybrid search (keyword + vector)** using **Azure AI Search**.
- Applied **metadata-based filtering** to retrieve department-specific documents.
- Used **Azure OpenAI embeddings** with **FAISS** for low-latency semantic retrieval.
- Built backend using **FastAPI** and frontend using **React.js**.
- Reduced internal query resolution time by **~60%**.

Conversational RAG System with Chat History:

- Implemented **conversation-aware RAG** using **LangChain Memory**.
- Enabled **follow-up question handling** via **query rewriting**.
- Stored and reused chat history to improve multi-turn answer accuracy.
- Integrated **streaming responses** using **Azure OpenAI**.
- Improved response relevance by **~40%** for conversational queries.

Education

B.Sc. (Mathematics) – S.P. College, V.B.S.P.U | 2017 | 59%
Intermediate – S.K.D.I. College, UP Board | 2014 | 72.8%
High School – S.K.D.I. College, UP Board | 2012 | 71.6%